

AAI-501: Introduction to Artificial Intelligence
Final Team Project — Proposal / Status Update
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TOPIC

Our team proposes to build and evaluate a machine learning system that classifies URLs as phishing (malicious) or legitimate based on a set of engineered structural and lexical features. Phishing remains one of the most common techniques for scams and data breaches, and automated detection is a practical, high-impact application of supervised learning.

TEAM STATUS

Our team has met and discussed several candidate datasets and problem areas, including demand forecasting and customer churn. After reviewing the option, we have decided on the phishing detection problem as the one we are most interested in classifying categorical, feature-based data and find the security application compelling. We have also agreed on a standing weekly meeting time to review progress and coordinate deliverables, and we will finalize a specific division of labor among team members in the coming week.

1. PROBLEM, ALGORITHMS, AND SYSTEM

We will use a publicly available phishing dataset (e.g., the UCI Phishing Websites dataset and/or the Mendeley phishing URL dataset), which contains tens of thousands of labeled examples with pre-engineered features, so heavy preprocessing is not required. We intend to build a classification pipeline and compare at least two families of algorithms — for example, Logistic Regression as an interpretable baseline against ensemble methods such as Random Forest and gradient boosting (XGBoost). We will tune model parameters and compare results empirically. The system will take URL features as input and output a phishing / legitimate prediction with a confidence score.

2. RELATED COURSE TOPICS

This project draws on supervised learning and binary classification, feature selection and variable importance, ensemble methods, and model evaluation (precision, recall, ROC/AUC). We may also incorporate a simple neural network to connect the work to deep learning concepts covered in the course.

3. EXPECTED SYSTEM BEHAVIOR

Given a set of URL features, the trained model should reliably flag malicious URLs while keeping false positives low. We expect the models to learn that certain features — such as the

presence of an IP address in the URL, unusual URL length, use of "@" symbols, or suspicious subdomains — are strong indicators of phishing.

4. ISSUES WE EXPECT TO FOCUS ON

Key issues include class imbalance and the choice of appropriate evaluation metrics beyond accuracy, model interpretability (identifying which features drive predictions), and testing whether a simpler model using only the most important features can match the full model's performance.

5. INITIAL RESOURCES (APA 7)

Mohammad, R. M., Thabtah, F., & McCluskey, L. (2014). Predicting phishing websites based on self-structuring neural network. *Neural Computing and Applications*, 25(2), 443-458.

Sahingoz, O. K., Buber, E., Demir, O., & Diri, B. (2019). Machine learning based phishing detection from URLs. *Expert Systems with Applications*, 117, 345-357.

Vrbancic, G., Fister, I., & Podgorelec, V. (2020). Datasets for phishing websites detection. *Data in Brief*, 33, 106438.